Michael Wray

Senior Lecturer in Computer Vision

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Current Appointment

Senior Lecturer in Computer Vision (Grade L) February 2026 – Current
University of Bristol *Bristol, UK*

Previous Appointments

Lecturer in Computer Vision (Grade K) August 2024 – February 2026
University of Bristol *Bristol, UK*

Lecturer in Computer Vision (Grade J) March 2022 – August 2024
University of Bristol *Bristol, UK*

Post-Doctoral Researcher in Computer Vision Dec. 2019 – March 2022
University of Bristol *Bristol, UK*

Teaching Associate Sept. 2018 – Sept. 2019
University of Bristol *Bristol, UK*

- COMS21202: Lead the restructure of labs and coursework for 2nd year Computer Science Unit Symbols, Patterns and Signals.

Teaching Assistant Sept. 2015 – Sept. 2019
University of Bristol *Bristol, UK*

Data Structures and Algorithms (Y2) Symbols, Patterns and Signals (Y2) Computer Graphics (Y3) Image Processing and Computer Vision (Y3) Applied Deep Learning (Y4)

Academic Qualifications

PhD in Computer Vision Sept. 2015 – Sept. 2019
University of Bristol *Bristol, UK*

- Thesis Title: Verbs and Me—An Investigation into Verbs as Labels for Action Recognition in Video Understanding.

MEng in Computer Science Sept. 2011 – Sept. 2015
University of Bristol *Bristol, UK*

- Degree Classification: First Class Honours
- Dissertation Title: Generating Object Proposals for Wearable Visual Sensors.

Awards and Honours

EgoVis 2024 Distinguished Paper Award *Egocentric Video-Language Pretraining*

EgoVis 2024 Distinguished Paper Award *Ego4D: Around the World in 3,000 Hours of Egocentric Video*

Best Paper Award Candidate CVPR 2022 (Top 0.1% of papers) *Ego4d: Around the world in 3,000 hours of egocentric video*

Charitable Donation - Google DeepMind *£50,000 awarded for my work on the EPIC-Kitchens Datasets*

European Laboratory for Learning and Intelligent Systems (ELLIS) Member *Accepted March 2022. Nominated by Jan Van Gemert (TU Delft) and Giovanni Maria Farinella (University of Catania)*

Outstanding Reviewer *In all major Computer Vision Conferences: BMVC 2024, ECCV 2022, ICCV 2021, CVPR 2021, BMVC 2020*

Best Poster Award (Honourable Mention) *British Machine Vision Association Summer School, Swansea 2016*

Invited Talks

BMVA Summer School, 2022-2025: *Egocentric Vision - Making Sense of the First-Person Perspective*

VIViD Research Seminar, Durham University, 2024: *Fine Grained Video Understanding from a Personal Perspective*

Advancements in Time Series Analysis for Computer Vision: Techniques, Applications, and Challenges, 2024: *Unlocking the Temporal Dimension from the Egocentric Perspective*

Data Collection at the Open Access Databases Conference, 2023: *Open Datasets in Computer Vision*

Video Understanding Symposium, 2022: *Do we still need Classification for Video Understanding?*

Research Lecture University of Catania, 2022 *Teaching Computers to Expand their Vocabulary*

Russell Group Teachers Conference, 2022 *From Access to Bristol Student to Lecturer*

BMVA Symposium: Robotics Meets Semantics, 2018: *Towards an Unequivocal Representation of Actions*

Egocentric Perception Interaction and Computer Workshop, 2016: *SEMBED: Semantic Embedding of Egocentric Action Videos*

Teaching and Related Admin

I currently teach on 1 unit as Unit Director and 1 unit as a lecturer with a mixture of undergraduate teaching covering years 2, and 4. Previously, I was the unit director of the 4 Individual Project units. I have updated the teaching for all of the units I am a part of, introducing new focuses of research-rich learning, video lecture content, and support sessions to improve the teaching experience for students. I consistently receive excellent feedback for my lecturing, teaching style, and organisation for my units. I am the Programme Director for the Computer Science with AI programme which had its first intake in the 2025/26 year and I am overseeing the creation of new units for the degree rpogram with a team of 5 lecturers. A particular focus of my Programme Director role has also been how AI can be integrated into the School of Computer Science's teaching and asssessments.

Unit Director: Applied Deep Learning COMSM0045 2023 – Current

- 66 Students 2022/23, 42 students 2023/24, 37 students 2024/25, 51 students 2025/26
- 4th Year Undergraduate Optional Unit.
- Responsibilities: 50% teaching, 50% exam, 100% Coursework, and Unit Director Admin.
- Examined Coursework and written exam.

- I continuously update lecture/lab material and lab/coursework content to ensure up to date teaching delivery for students wishing to work in this area. This has included 2 new lectures and 1 new lab since I joined as UD.
- I introduced several teaching and assessment methods: **Practical Exam Questions**, **GitLab Integration** (see below).

Lecturer: Computer Systems A COMS20008 2023 – Current

- 165 Students 2022/23, 148 students 2023/24, 199 students 2024/25, 172 students 2025/26.
- 2nd Year Undergraduate Core Unit.
- Responsibilities: 40% teaching, 50% exam, 20% Coursework.
- Examined Coursework and multiple choice exam.
- I teach the Distributed Systems side of the unit through recorded lectures, labs, and Q&A sessions.
- I introduced a new lecture to help with the Distributed Component of the coursework and new Q&A materials for the coursework/exam/components
- I have created high quality lecture videos that are engaging for students to watch and learn from due to the asynchronous nature of the unit.

Unit Director 23/24: Individual Project Units 2023 – 2025

Deputy Unit Director 22/23

- COMS30044, COMS30045, COMSM0052, COMSM0142.
- 157 Students 2022/23, 167 students 2023/24, 119 students 2024/25.
- 3rd/4th Year Undergraduate Core Unit. Capstone project unit, 100% coursework, Dissertation and viva
- I was the unit director for the individual project units and oversaw admin tasks such as matching of students/project/supervisors, marking deadlines and viva dates, ethics training, organisation of research skill workshops.
- I introduced a new ethics lecture and quiz, resulted in a reduction of ethical oversights for 22/23.
- I have organised writing retreats new to the course to help with student's time management, planning, and writing with good feedback and attendance.
- I organised for three years in a row a Poster Day for students to received formative feedback and showcase their work to other undergraduate students/staff. This was the first time this was run since before the pandemic requiring new policies and expectations for staff and students.
- I became the acting Unit Director during the 2023 Marking and Assessment Boycott. I stepped up and lead the marking process during the tumultuous time within moderation panels and adminstering emergency markers over the reassessment period.
- I introduced a new AI policy and explored the use of AI inside these Capstone projects: **AI Usage in Individual Projects** (see below).

Innovative Teaching and Assessments

- **AI Coursework Investigation:** As part of the School Education Committee, I led the investigation into how vulnerable current 3rd and 4th year coursework and Independent Learning Objectives are to malicious/forbidden AI usage. PhD students were hired to complete 3rd year courseworks using only AI prompts. The investigation highlighted certain areas to improve and will dictate future school policy on assessment creation.
- **AI Usage in Individual Projects:** Created a new policy on how AI could be used in the project units: students were able to use AI for anything but writing up the dissertation. Updates to the mark scheme required AI usage to be justified as well as a mandatory appendix to provide details as to what AI tool was used and why. This culminated into an investigation along the lines of what students used AI for and how this effects their mark. The main takeaways were that 75% of students used AI in some capacity for coding and LaTeX help as the highest use-cases. There was found to be no correlation between AI usage and final grade achieved. The external examiner heavily praised the policy as exactly what every university should be doing in this changing landscape.

- **GitLab Integration for Applied Deep Learning:** Working with the Tech Services team to integrate GitLab into formative and summative coursework submission and marking. Created new formative assigments for the students testing their knowledge so far on both theoretical and practical aspects of the course. These answers are then submitted via a GitLab repository that has been created per student and then marked with feedback automatically being placed within their repositories. This has the benefit for summative coursework to check how the student progresses with the assignment as another academic integrity check whilst also providing more feedback options both pre and post assignment. With the successful pilot, this will be rolled out to more courses in the School of Computer Science in the future.
- **Practical Exam Questions:** I included applied questions in the exam so that the labs match learning objectives with the restructure of units into major and minor versions for the 2024/25 year. Students have provided good feedback to these essay style questions and enjoyed the practical required.

Postgraduate Advising

- Rachael Laidlaw: PhD, 2025–Current *primary supervisor* (w/ Edwin Simpson and Raul Santos-Rodriguez)
- Fahd Abdelazim: PhD, 2024–Current *primary supervisor*
- Sam Pollard: MEng, PhD, 2023–Current *primary supervisor*
- Beth Pearson: PhD, 2023–Current *primary supervisor* (w/ Martha Lewis)
- Shijia Feng: PhD, 2022–2025 – submitted (w/ Walterio Mayol Cuevas)
- Kevin Flanagan: PhD, 2021–2025 – defended *primary supervisor* (w/ Dima Damen)
- Adriano Fragomeni: PhD, 2020–2025 – defended (w/ Dima Damen)

Professional Development

I have completed both PGCAP units, unit 1 in 2023 and unit 2 in 2024. I have also attended on BILT workshops including Programme Director Induction, AI and Assessment, Group Assessment, Personal Tutoring, and Research Supervision Courses.

Collaborative Teaching Projects

External In collaboration with the British Machine Vision Association, a national forum for researchers in machine vision, image processing, and pattern recognition within the UK, I have been invited to give a 90 minute lecture on egocentric vision. This event is attended by approximately 60 1st year PhD students from UK and abroad. **Internal** I was a critical friend for the staff-facing AI and Education Module developed by Aisling Tierney as part of BILT. I provided feedback across the different learning materials and modules.

Research and Related Admin

I am currently supervising 4 PhD students, of which I am the primary supervisor for all students. Three of the students I have supervised in the past have submitted their thesis with two having already defended and undergoing corrections. I have six papers at CVPR, Computer Vision's premier conference (note that in Computer Vision conferences are ranked as high as the top journals). Two of these were a collaboration with Meta and 15 Universities worldwide and two in collaboration with Czech Technical University in Prague. In total, I have 23 papers across Computer Vision's and Machine Learning's top conferences and journals. One of my papers was a best paper award candidate (top 0.1% of papers) at CVPR 2022 and two of my papers have been awarded the distinguished paper award at the annual EgoVis workshop, a prize that recognises papers that have had a long-term impact on egocentric vision. Due to my expertise in this area, I have been an area chair for 6 conferences and will be a Lead Area Chair for CVPR 2026 as well as being awarded a charitable donation from Google DeepMind (£50,000). Finally, I have been invited and presented a lecture at the British Machine Vision Association's Summer

School for 4 years in a row, and have been invited as an external examiner for 3 PhD Theses. Orcid ID: https://orcid.org/0000-0001-5918-9029

Publications

Academic Journal Papers (Refereed)

- Feng, S., Wray, M., Sullivan, B., Jang, Y., Ludwig, C., Gilchrist, I., and Mayol-Cuevas, W., 2025. Are you struggling? dataset and baselines for struggle determination in assembly videos. *International Journal of Computer Vision, 1-38.* Springer International Publishing. **Contribution:** Defined dataset tasks; paper writing; and figure creation.
- Grauman, K., Westbury, ... and Wray, M. and ... (17th of 85 **total** authors), 2022. Ego4d: Around the world in 3,000 hours of egocentric video. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2024. Institute of Electrical and Electronics Engineers. **Contribution:** Data collection lead for Bristol site; co-developed taxonomy of annotations; formulation of benchmark Episodic Memory tasks; dataset construction; and paper writing.
- Damen, D., Doughty, H., Farinella, G.M., Furnari, A., Kazakos, E., Ma, J., Moltisanti, D., Munro, J., Perrett, T., Price, W. and Wray, M., 2022. Rescaling egocentric vision: Collection, pipeline and challenges for epic-kitchens-100. *International Journal of Computer Vision, 1-23.* Springer International Publishing. **Contribution:** Data collector for dataset; defined dataset tasks; lead the parsing and clustering of narrations; paper writing and figure creation; lead the dataset release on Github.
- Damen, D., Doughty, H., Farinella, G.M., Fidler, S., Furnari, A., Kazakos, E., Moltisanti, D., Munro, J., Perrett, T., Price, W. and Wray, M., 2020. The epic-kitchens dataset: Collection, challenges and baselines. *IEEE Transactions on Pattern Analysis and Machine Intelligence, 43(11), 4125-4141.* Institute of Electrical and Electronics Engineers. **Contribution:** Data collector for dataset; defined Multi-Instance Retrieval Task and ran baselines; lead for dataset task challenges; lead the parsing and clustering of narrations; paper writing and figure creation; lead the dataset release on Github.

Conference Contributions (Refereed)

- Pearson, B., Boulbarss, B., Wray, M., and Lewis, M., 2025. Evaluating Compositional Generalisation in VLMs and Diffusion Models. *In Proceedings of the Joint Conference on Lexical and Computational Semantics (*SEM), 14*, 2025. Special Interest Group on the Lexicon of the Association for Computational Linguistics. **Contribution:** Idea development; paper writing.
- Flanagan, K., Damen, D. and Wray, M., 2025. Moment of Untruth: Dealing with Negative Queries in Video Moment Retrieval. *In Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 12*, 2025. Institute of Electrical and Electronics Engineers. **Contribution:** Idea formulation and development; paper writing; project management and synthesis – last author.
- Pollard, S. and Wray, M., 2025. Video, How do your tokens merge?. *In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, 38, 2025. Institute of Electrical and Electronics Engineers. **Contribution:** Idea Development, Method Development, Paper Writing, and project management and synthesis – last author.
- Perrett, T., Darkhalil, A., Sinha, S., Emara, O., Pollard, S., Parida, K. K., ... and Wray, M., and ..., Damen, D., (17th of **19** authors), 2025. HD-EPIC A highly detailed egocentric video dataset. *In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 38, 2025. Institute of Electrical and Electronics Engineers. **Contribution:** Data collector; Idea development; Parsing team lead; Human evaluation; Annotation checking and quality control; Benchmark development; and Paper writing.
- Soucek, T., Gatti, P., Wray, M., Laptev, I., Damen, D., Sivic, J., 2025. Showhowto: Generating scene-conditioned step-by-step visual instructions. *In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 38, 2025. Institute of Electrical and Electronics Engineers. **Contribution:** Methodology development, related work, human evaluation, and paper writing.

- Mucha, W., Wray, M., Kampel, M., 2024. SHARP: Segmentation of hands and arms by range using pseudo-depth for enhanced egocentric 3D hand pose estimation and action recognition. *In Proceedings of the IAPR International Conference on Pattern Recognition*, 27, 2024. International Association for Pattern Recognition. **Contribution:** Initial idea formulation, methodology development, and paper writing.
- Zhu, B., Flanagan, K., Fragomeni, A., Wray, M., Damen, D., 2024. Video editing for video retrieval. *In Proceedings of the Springer European Conference on Computer Vision Workshops*, 18, 2024. Springer International Publishing. **Contribution:** Idea and methodology development, figure creation, and paper writing.
- Souček, T., Damen, D., Wray, M., Laptev, I. and Sivic, J., 2024. Genhowto: Learning to generate actions and state transformations from instructional videos. *In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 37, 2024. Institute of Electrical and Electronics Engineers. **Contribution:** Idea and methodology development, background/related work; and paper writing.
- Grauman, K., Westbury, A., Torresani, L., Kitani, K., Malik, J., Afouras, T., Ashutosh, K., Baiyya, V., Bansal, S., Boote, B. and Byrne, E. and ... and Wray, M. (101st of **101** total authors), 2024. Ego-exo4d: Understanding skilled human activity from first-and third-person perspectives. *In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 37, 2024*. Institute of Electrical and Electronics Engineers. **Contribution:** definition of narration guidelines and questionnaire collection; definition of the Procedure Understanding task; and paper writing/figure creation.
- Flanagan, K., Damen, D. and Wray, M., 2023. Learning temporal sentence grounding from narrated egovideos. *In Proceedings British Machine Vision Conference (BMVC), 34*, 2023. British Machine Vision Association. **Contribution:** Idea formulation and development; paper writing; project management and synthesis – last author.
- Lin, K.Q., Wang, J., Soldan, M., Wray, M., Yan, R., Xu, E.Z., Gao, D., Tu, R.C., Zhao, W., Kong, W. and Cai, C., 2022. Egocentric video-language pretraining. *In Proceedings Advances in Neural Information Processing Systems, 35, 7575-7586.* Curran Associates. **Contribution:** Input on Experimental design and paper writing.
- Fragomeni, A., Wray, M. and Damen, D., 2022. ConTra:(Con) text (Tra) nsformer for cross-modal video retrieval. *In Proceedings of the Asian Conference on Computer Vision, 16, 3481-3499.* Springer International Publishing. **Contribution:** Idea and methodology development, paper writing/figure creation.
- Grauman, K., Westbury, A., Byrne, E., Chavis, Z., Furnari, A., Girdhar, R., Hamburger, J., Jiang, H., Liu, M., Liu, X. and Martin, M. ... and Wray, M. and ..., (17th of 85 **total** authors) 2022. Ego4d: Around the world in 3,000 hours of egocentric video. *In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 35, 18995-19012* Institute of Electrical and Eloctronics Engineers. **Contribution:** Data collection lead for Bristol site; co-developed taxonomy of annotations; formulation of benchmark Episodic Memory tasks; dataset construction; and paper writing.
- Wray, M., Doughty, H. and Damen, D., 2021. On semantic similarity in video retrieval. *In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 34, 3650-3660.* Institute of Electrical and Electronics Engineers. **Contribution:** Initial idea formulation and development; experiments on EPIC-Kitchens dataset; paper writing and figure creation; code, challenge, and dataset release.
- Wray, M. and Damen, D., 2019. Learning visual actions using multiple verb-only labels. *In Proceedings of the British Machine Vision Conference (BMVC), 30, 2019.* British Machine Vision Association. **Contribution:** Idea development; ran all experiments; collected survey results from Amazon Mechanical Turk; paper writing and figure creation; dataset and code release.
- Wray, M., Larlus, D., Csurka, G. and Damen, D., 2019. Fine-grained action retrieval through multiple parts-of-speech embeddings. *In Proceedings of the IEEE/CVF international conference on computer vision, 17, 450-459.* Institute of Electrical and Electronics Engineers. **Contribution:** Initial idea formulation and development; code writing and experimental testing for entire project; paper writing and figure creation; code and dataset open release—first author.

- Damen, D., Doughty, H., Farinella, G.M., Fidler, S., Furnari, A., Kazakos, E., Moltisanti, D., Munro, J., Perrett, T., Price, W. and Wray, M., 2018. Scaling egocentric vision: the epic-kitchens dataset. *In Proceedings of the European conference on computer vision (ECCV), 15, 720-736.* Springer International Publishing. **Contribution:** Data collector for dataset; defined dataset tasks; lead the parsing and clustering of narrations; paper writing and figure creation; lead the dataset release on Github.
- Moltisanti, D., Wray, M., Mayol-Cuevas, W. and Damen, D., 2017. Trespassing the boundaries: Labeling temporal bounds for object interactions in egocentric video. In Proceedings of the IEEE International Conference on Computer Vision, 16, (pp. 2886-2894). Institute of Electrical and Electronics Engineers. **Contribution:** Ran human study on Amazon Mechanical Turk and contributed to paper editing.
- Wray, M., Moltisanti, D., Mayol-Cuevas, W. and Damen, D., 2016. Sembed: semantic embedding of egocentric action videos. *In Computer Vision–ECCV 2016 Workshops: Amsterdam, The Netherlands, October 8-10 and 15-16, 2016, Proceedings, Part I 14 532-545.* Springer International Publishing. **Contribution:** Initial idea formulation; code writing and experimental testing for proposed method; figure creation; paper writing—joint 1st author

Forthcoming Publications

Conference Contributions (Refereed)

- Fragomeni, A., Damen, D. and Wray, M., 2025. Leveraging Modality Tags for Enhanced Cross-Modal Video Retrieval. *In Proceedings British Machine Vision Conference (BMVC), 36*, 2025. British Machine Vision Association. **Contribution:** Idea development; paper writing; project management and synthesis – last author.

Research Grants and Funding

- Automated image analysis to facilitate the incorporation of quality assurance measures into surgical RCTs, *Elizabeth Blackwell and Jean Golding Institutes' AI in Health Award* £3,000, 100% Co-PI, 2025–2026
- EPIC-Kitchens Dataset, Charitable Donation, *Google DeepMind*, £50,000, 100% PI, 2022

Indications of External Recognition

See above for invited talks.

- Associate Editor Special Issue on Text Multimedia Retrieval: retrieving multimedia data by means of natural language 2024.
- Associate Editor TheIET Computer Vision Journal 2024–Current.
- Member of European Laboratory for Learning and Intelligent Systems 2022–Current
- Member of Ego4D Consortium Faculty 2022–Current
- Area Chair of major Computer Vision/Machine Learning Conferences: CVPR 2023/24/25/26, ECCV 2024, NeurIPS 2024/25.
- Internal PhD Examiner for 2 theses: Ruixiong Wang, 2024; Mengjie Zhou, 2024.
- External PhD Examiner for 3 theses: *Tanqiu Qiao*, **Durham University, 2024**. *Edward Fish*, **University of Surrey, 2024**. *Ivan Rodin*, **University of Catania, 2023**.
- External MScR Examiner: Jack Tsangou, University of Edinburgh 2023.

Related Administration

- **Chair of BMVA Research Symposium Jan 2024** Organising chair of a one-day reseach event on Vision and Language Understanding in Collaboration with the BMVA of speakers from UK and Europe with 90 attendees from UK and Europe. This was the second such event since the pandemic and the first event that brought together the Vision and Language research community in UK and Europe.
- **Collaboration with Project ARIA from Meta** I am Co-I working with the Project ARIA research team within Meta's Reality Labs on the Project ARIA research devices to record new datasets for egocentric vision. These include charitable donations of 18 devices.

Academic Leadership and Citizenship

As Programme Director for Computer Science with AI, I have provided strategic leadership in curriculum development and quality assurance. I am overseeing the creation of the programme's new units, leading a diverse group of academics to ensure an inclusive and balanced approach that integrates both new and experienced staff. I am a member of the University of Bristol's AI Leadership group as well as the DRI User Executive working on providing school and university level solutions to AI and research infrastructure. Due to my expertise, I have been a regular reviewer for conferences, an examiner for PhDs (3 external, 2 internal) as well as a regular workshop organiser, promoting both the research area and my work.

- Member of the University of Bristol's AI Leadership Group since 2025.
- Member of the DRI User Executive (re-naming of HPC Executive) since 2024.
- Member of the Advanced Computer Research Centre High Performance Computer Executive 2023-2024.
- Member of Working Group for School of Computer Science Research Leave 2024.
- Regular Reviewer of **major Computer Vision Conferences**: CVPR 2019–2022, , ICCV 2021–2023, ECCV 2022, ACCV 2020–2022, BMVC 2019–2024, WACV 2021–2026 and **major Machine Learning Conferences**: AAAI 2025, ICLR 2025, ICML 2024, ICRA 2025, NeurIPS 2023
- Regular Reviewer of major Computer Vision Journals: TPAMI, IJCV, TCSVT, Pattern Recognition.
- Lead Area Chair for CVPR 2026
- Co-organiser of workshops at major Computer Vision Conferences: What is Next in Video Understanding? (CVPR2024); Egocentric Perception, Interaction and Computing al(ICCV2021–lead, CVPR2021–lead, and ECCV2020); Joint Ego4D and EPIC workshop (CVPR 2022/2023); and Ego4D (ECCV 2022).
- Academic lead of the Formula Student-AI team at University of Bristol—creating workshops and teaching students across the Engineering faculty fundamentals of Machine Learning and AI.
- Always meet internal marking deadlines for coursework and exam marking, liaising frequently with the school office as required.
- Volunteer lectures for other units, including Intro to Computer Science for the MSc Computer Science Conversion and Foundations of AI for the Practice-Oriented Artificial Intelligence CDT.